JNTUA R23 REGULATIONS

CONCRETE TECHNOLOGY LABORATORY

(23A01402P)

Laboratory Manual with Detailed Inputs, Procedures, Calculations and Outputs

Programme	Year / Semester	L	T	P	C
B.Tech. Civil Engineering	II Year B.Tech. CE – II Semester	0	0	3	1.5

Course Objectives

To test basic properties of ingredients of concrete, fresh concrete and hardened concrete properties.

Course Outcomes

CO	Outcome	Covered in
CO1	Outline importance of testing cement and its properties	Experiments 1.1 – 1.6
CO2	Assess different properties of Aggregates	Experiments 2.1 – 2.4, 3.1 – 3.3
CO3	Assess fresh concrete properties and their relevance to hardened concrete	Experiments 4.1 – 4.3
CO4	Assess hardened concrete properties	Experiments 5.1 – 5.5

How to read each experiment

Every experiment follows the same layout: **1 Aim → 2 Inputs** (apparatus, materials, quantities) **→ 3 Procedure → 4 Observations → 5 Calculations → 6 Outputs → 7 Precautions**. Observation tables contain *illustrative sample data* so that the full calculation chain and the expected form of the output can be seen; all sample values were calculated consistently and fall within typical ranges for Ordinary Portland Cement, Zone II river sand and 20 mm crushed granite. Acceptance limits quoted are from the relevant Bureau of Indian Standards (BIS) codes and should be verified against the latest edition of the code before submission.

Contents

No.	Experiment	Standard
1.1	Normal consistency of cement	IS 4031 (Part 4)
1.2	Fineness of cement	IS 4031 (Part 1)
1.3	Initial and final setting time of cement	IS 4031 (Part 5)
1.4	Specific gravity of cement	IS 4031 (Part 11)
1.5	Soundness of cement	IS 4031 (Part 3)
1.6	Compressive strength of cement	IS 4031 (Part 6)
2.1	Grading & fineness modulus of fine aggregate (sieve analysis)	IS 2386 (Part 1), IS 383
2.2	Specific gravity of fine aggregate	IS 2386 (Part 3)
2.3	Water absorption of fine aggregate	IS 2386 (Part 3)
2.4	Bulking of sand	IS 2386 (Part 3)
3.1	Grading of coarse aggregate (sieve analysis)	IS 2386 (Part 1), IS 383
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4.1	Workability — compaction factor test	IS 1199
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5.1	Compressive strength of concrete	IS 516
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5.4	Split tensile strength of concrete	IS 5816
5.5	Non-destructive testing (demonstration)	IS 13311 (Parts 1 & 2)
A	Appendix — Summary of acceptance limits, general lab safety	—

Experiment 1.1: Normal Consistency of Cement

Course Outcome: CO1 | **Reference standard:** IS 4031 (Part 4) – Vicat apparatus

1. Aim

To determine the percentage of water required to produce a cement paste of standard (normal) consistency. This percentage (P) is used to fix the water content for the setting-time, soundness and strength tests.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Cement	300 g OPC per trial, passed through 850 µm sieve
Water	Clean potable water, starting at 26% of cement mass (78 mL), increased in steps of 2%
Vicat apparatus	Plunger 10 mm dia × 50 mm long, moving mass 300 g, graduated scale 0–40 mm
Vicat mould	Conical, 80 mm top dia, 90 mm bottom dia, 40 mm high, on non-porous glass base plate
Others	Balance (1 g), measuring cylinder 200 mL, non-absorbent mixing tray, trowel, stop-watch; room 27 ± 2 °C

3. Procedure

1. Weigh 300 g of cement and place it on the tray; make a crater and add the measured water (start 26%).
2. Mix quickly with a trowel; gauging time must be 3–5 min, counted from the instant water is added.
3. Fill the Vicat mould (resting on the glass plate) with paste, shake slightly to expel air, and level the top with a trowel.
4. Place the mould under the plunger, lower the plunger gently until it touches the paste surface, tighten the clamp screw, and note the scale reading.
5. Release the plunger quickly and let it sink freely; read the penetration after 30 s as the distance of the plunger tip from the bottom of the mould.
6. Repeat with fresh paste, increasing the water by 2% each trial, until the plunger stops 5–7 mm from the bottom of the mould (penetration 33–35 mm).

4. Observations — Penetration vs water content

Sample readings are illustrative; students must replace them with their own laboratory readings.

Trial	Cement (g)	Water (% of cement)	Water (mL)	Plunger stops at (mm from bottom)	Penetration (mm)
1	300	26	78.0	15	25
2	300	28	84.0	11	29
3	300	30	90.0	6	34
4	300	32	96.0	3	37

5. Calculations

Normal consistency P (%) = (Mass of water added ÷ Mass of cement) × 100

Trial 3: P = (90 g ÷ 300 g) × 100 = 30% (plunger stops at 6 mm from bottom — within the 5–7 mm window)

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Normal consistency, P	30 %	Typical OPC range 26–33%
Water for 300 g cement	90 mL	= 0.30 × 300
Water for subsequent tests	0.85 P = 25.5 %	Used in Exp. 1.3 (setting time)

Result / Acceptance criteria

Normal consistency of the given cement = **30%**. The paste is of standard consistency when the plunger penetrates to a point 5–7 mm from the bottom of the mould.

7. Precautions

- Do not exceed the 3–5 min gauging time; paste must be fresh for every trial.
- Keep the Vicat apparatus free from vibration; clean the plunger after each trial.
- Wipe the mould and glass plate before refilling; use clean cement and water at 27 ± 2 °C.

Experiment 1.2: Fineness of Cement (Dry Sieving)

Course Outcome: CO1 | **Reference standard:** IS 4031 (Part 1)

1. Aim

To determine the fineness of cement by dry-sieving through a 90 µm IS sieve. Fineness controls the rate of hydration, strength gain and heat of hydration.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Cement	100 g OPC, air-dry, lumps broken by hand
Sieve	90 µm IS sieve with lid and pan
Balance	Accuracy 0.1 g (or 10 mg for higher precision)
Others	Soft brush, tray

3. Procedure

1. Weigh 100 g of cement (W_1) and place it on the clean, dry 90 µm sieve; break up lumps with fingers.
2. Agitate the sieve by swirling, planetary and linear movements continuously for 15 min.
3. Weigh the residue left on the sieve (W_2).
4. Repeat the test with a fresh sample and take the mean (agree within 1%).

4. Observations — Weighing

Sample readings are illustrative; students must replace them with their own laboratory readings.

Trial	Mass of cement W1 (g)	Residue on 90 µm W2 (g)	Fineness = W2/W1 × 100 (%)
1	100	7.5	7.5
2	100	7.9	7.9
Mean			7.7

5. Calculations

Residue (%) = (W_2 ÷ W_1) × 100

Mean residue = (7.5 + 7.9) ÷ 2 = 7.7%

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Residue on 90 µm sieve	7.7 %	Limit for OPC ≤ 10%
Fineness status	Satisfactory	Passes IS 269

Result / Acceptance criteria

Residue of the given cement on the 90 µm sieve = **7.7% (≤ 10%)** – cement is acceptable as regards fineness.

7. Precautions

- Sieve must be dry and clean; do not tap or press cement through the mesh.
- Avoid sieving in a humid atmosphere (cement forms lumps).
- Weigh residue immediately after sieving.

Experiment 1.3: Initial and Final Setting Time of Cement

Course Outcome: CO1 | **Reference standard:** IS 4031 (Part 5) – Vicat apparatus

1. Aim

To determine the initial and final setting times of cement paste of normal consistency.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Cement	300 g OPC
Water	0.85 × P of cement mass = 0.85 × 30% = 25.5% → 76.5 mL
Vicat needle (initial)	1 mm square section (≈1.13 mm dia), 50 mm long; moving mass 300 g
Annular attachment (final)	Needle with 5 mm dia metal annulus projecting 0.5 mm beyond needle tip
Others	Vicat mould, glass plate, stop-watch, trowel, humidity chamber (RH ≥ 90%, 27 ± 2 °C)

3. Procedure

1. Prepare paste with 0.85 P water; start the stop-watch the moment water is added to cement (t = 0).
2. Fill the Vicat mould, level, and place it under the Vicat needle; keep the mould in the humid chamber between readings.
3. Lower the needle gently onto the paste, release, and read the distance of the needle tip from the bottom of the mould.
4. Repeat every 10–15 min (cleaning the needle each time) at different spots at least 5 mm from previous holes and the mould wall.
5. **Initial setting time** = time from water addition until the needle stays 5.0 ± 0.5 mm from the bottom of the mould.
6. Replace the needle with the annular attachment; **final setting time** = time until the needle makes an impression but the annular ring fails to.

4. Observations — Needle penetration vs time

Sample readings are illustrative; students must replace them with their own laboratory readings.

Time from adding water (min)	30	60	90	105	120	…	270
Needle tip from bottom (mm)	0.5	1.8	3.4	4.2	5.0	…	Needle marks, ring does not

5. Calculations

Water = 0.85 × P × 300 g ÷ 100 = 0.85 × 30 × 3 = 76.5 g

Initial setting time = time at which needle stops 5.0 ± 0.5 mm from base = 120 min

Final setting time = time when annular ring just fails to leave an impression = 270 min (4 h 30 min)

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Initial setting time	120 min	OPC limit ≥ 30 min
Final setting time	270 min	OPC limit ≤ 600 min
Water used	76.5 mL	0.85 P

Result / Acceptance criteria

Initial setting time = **120 min** (≥ 30 min) and final setting time = **270 min** (≤ 600 min); the cement conforms to IS 269 / IS 1489 requirements for OPC.

7. Precautions

- Never allow the needle to be dropped; release it gently.
- Test place must be free of vibration; keep the paste in a humid environment.
- Clean the needle after each reading to avoid wrong results.

Experiment 1.4: Specific Gravity of Cement

Course Outcome: CO1 | **Reference standard:** IS 4031 (Part 11) – Le Chatelier flask

1. Aim

To determine the specific gravity (relative density) of Ordinary Portland Cement using the liquid-displacement method.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Cement	64 g OPC (dry)
Liquid	Kerosene (or naphtha), free from water
Le Chatelier flask	250 mL flask with graduated neck (0–1 mL and 18–24 mL marks)
Others	Balance (0.01 g), funnel, glass rod, thermostatic water bath at 27 ± 1 °C

3. Procedure

1. Fill the dry flask with kerosene up to a mark between 0 and 1 mL; immerse in the water bath and record the initial reading V_1.
2. Weigh 64 g of cement (W) and introduce small quantities at a time through the funnel without splashing, avoiding any lump formation.
3. Roll the flask in an inclined position until no air bubbles rise to the surface.
4. Place the flask again in the water bath and record the final level reading V_2.
5. Repeat with another 64 g sample; the two results must agree within 0.01.

4. Observations — Displacement readings

Sample readings are illustrative; students must replace them with their own laboratory readings.

Trial	Mass of cement W (g)	Initial reading V1 (mL)	Final reading V2 (mL)	Volume displaced V2 − V1 (mL)	Specific gravity
1	64	0.4	20.7	20.3	3.153
2	64	0.5	20.8	20.3	3.153

5. Calculations

Specific gravity G = Mass of cement ÷ Volume of kerosene displaced = W ÷ (V_2 − V_1)

G = 64 ÷ 20.3 = 3.153 ≈ 3.15

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Specific gravity of cement	3.15	Typical OPC 3.10–3.15
Density of cement	3150 kg/m³	= G × 1000

Result / Acceptance criteria

Specific gravity of the given OPC = **3.15**. This value is used in mix design to compute the absolute volume of cement.

7. Precautions

- Flask and kerosene must be perfectly dry; any moisture will hydrate the cement.
- Introduce cement slowly to avoid air entrapment and splashing.
- Record both readings at the same bath temperature.

Experiment 1.5: Soundness of Cement (Le Chatelier Method)

Course Outcome: CO1 | **Reference standard:** IS 4031 (Part 3)

1. Aim

To determine the soundness (volume stability) of cement by measuring the expansion caused by free lime and magnesia, using the Le Chatelier apparatus.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Cement	100 g OPC
Water	0.78 P = 0.78 × 30 = 23.4% → 23.4 mL
Le Chatelier mould	Split brass cylinder 30 mm dia × 30 mm high, two indicator arms 165 mm long
Others	Two glass plates, 50 g weight, water bath, boiling arrangement, oil, scale (0.5 mm)

3. Procedure

1. Oil the mould lightly and place it on an oiled glass plate; mix 100 g cement with 0.78 P water.
2. Fill the mould, keeping the edges of the split together; cover with another oiled glass plate and a 50 g weight.
3. Submerge the mould in water at 27 ± 2 °C for 24 h; measure the distance between the indicator points (A).
4. Bring the water to boiling in 25–30 min and keep it boiling for 3 h (180 ± 5 min); allow it to cool.
5. Remove the mould and again measure the distance between the pointers (B). Expansion = B − A.

4. Observations — Indicator separation

Sample readings are illustrative; students must replace them with their own laboratory readings.

Specimen	Distance after 24 h water curing, A (mm)	Distance after 3 h boiling, B (mm)	Expansion B − A (mm)
1	18.0	20.5	2.5
2	17.5	19.5	2.0
Mean			2.25

5. Calculations

Expansion (mm) = B − A

Mean expansion = (2.5 + 2.0) ÷ 2 = 2.25 mm

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Mean expansion	2.25 mm	Limit for OPC ≤ 10 mm
Soundness	Sound	Free lime within limits

Result / Acceptance criteria

Le Chatelier expansion of the cement = **2.25 mm (≤ 10 mm)**; the cement is sound.

7. Precautions

- Avoid pushing the indicator arms apart while handling the mould.
- Keep boiling time exactly 3 h.
- Handle hot mould with tongs; allow to cool before measuring.

Experiment 1.6: Compressive Strength of Cement (Mortar Cubes)

Course Outcome: CO1 | **Reference standard:** IS 4031 (Part 6); IS 269 (grade requirements)

1. Aim

To determine the compressive strength of cement at 3, 7 and 28 days using 70.6 mm standard mortar cubes (cement : standard sand = 1 : 3).

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Cement / sand	Per cube: 200 g cement + 600 g standard (Ennore) sand → 3 cubes per age = 9 cubes
Water	(P/4 + 3)% of combined mass = (30/4 + 3) = 10.5% × 800 g = 84 mL per cube
Cube moulds	70.6 mm × 70.6 mm × 70.6 mm (face area ≈ 50 cm²), with base plate
Equipment	Vibrating machine (12000 ± 400 vibrations/min), trowel, balance, curing tank at 27 ± 2 °C
Testing machine	Compression testing machine; loading rate 35 N/mm² per min (≈ 175 kN/min on 50 cm²)

3. Procedure

1. Thoroughly mix 200 g cement and 600 g sand dry for one minute; add (P/4 + 3)% water and mix for 3–4 min until uniform colour.
2. Fill the greased mould immediately and vibrate for 2 min at the specified speed; level and cover with a glass plate.
3. Keep the moulds at 27 ± 2 °C and ≥ 90% RH for 24 h; demould and cure the cubes in clean water.
4. Test three cubes each at 3 days, 7 days and 28 days (surface-dry), loaded on faces which were in contact with the mould sides.
5. Apply load axially, uniformly at 35 N/mm² per min, and record the failure load; compute strength as load ÷ area.

4. Observations — Failure loads and strength (load in kN → strength in N/mm² on 5000 mm²)

Sample readings are illustrative; students must replace them with their own laboratory readings.

Age (days)	Cube 1	Cube 2	Cube 3	Mean (N/mm²)
3	125 kN → 25.0	128 kN → 25.6	122 kN → 24.4	25.0
7	172 kN → 34.4	168 kN → 33.6	176 kN → 35.2	34.4
28	225 kN → 45.0	219 kN → 43.8	231 kN → 46.2	45.0

5. Calculations

Compressive strength = Failure load (N) ÷ Area of cube face (mm²) = P ÷ 5000

3 days: 125 000 ÷ 5000 = 25.0 N/mm² (and so on for other cubes)

Mean: 3 d = 25.0, 7 d = 34.4, 28 d = 45.0 N/mm²

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
3-day strength	25.0 N/mm²	OPC 43: ≥ 23 (33 grade ≥ 16; 53 grade ≥ 27)
7-day strength	34.4 N/mm²	OPC 43: ≥ 33 (33 grade ≥ 22; 53 grade ≥ 37)
28-day strength	45.0 N/mm²	OPC 43: ≥ 43 (33 grade ≥ 33; 53 grade ≥ 53)

Result / Acceptance criteria

The cement satisfies the requirements of **43-grade OPC** (IS 269): 3 d = 25.0, 7 d = 34.4, 28 d = 45.0 N/mm².

7. Precautions

- Mix water must be measured accurately; use only standard sand.
- Keep cubes always moist before testing; test in surface-dry condition.
- Place cube so that the load acts perpendicular to the casting direction.

Experiment 2.1: Grading and Fineness Modulus of Fine Aggregate (Sieve Analysis)

Course Outcome: CO2 | **Reference standard:** IS 2386 (Part 1); IS 383

1. Aim

To determine the particle-size distribution, fineness modulus and grading zone of the given fine aggregate (sand).

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Sample	500 g sand, oven-dried at 105–110 °C and cooled
IS sieves	10 mm, 4.75 mm, 2.36 mm, 1.18 mm, 600 µ, 300 µ, 150 µ and pan (with lid)
Equipment	Mechanical sieve shaker (or manual), balance (0.1 g), tray, brush

3. Procedure

1. Take 500 g of dry sand and arrange the sieves in descending order of size (10 mm on top, pan at the bottom).
2. Place the sample on the top sieve, cover it and shake on the mechanical shaker for 10 min.
3. Weigh the mass retained on each sieve and on the pan; clean the mesh with a brush (do not push particles through).
4. Check that the sum of masses equals the original within 0.5%.
5. Calculate the percentage retained, cumulative percentage retained and percentage passing; plot the grading curve.

4. Observations — Sieve analysis (sample mass = 500 g)

Sample readings are illustrative; students must replace them with their own laboratory readings.

IS sieve	Mass retained (g)	% retained	Cumulative % retained	% passing
10 mm	0	0.0	0.0	100.0
4.75 mm	15	3.0	3.0	97.0
2.36 mm	40	8.0	11.0	89.0
1.18 mm	90	18.0	29.0	71.0
600 µ	130	26.0	55.0	45.0
300 µ	140	28.0	83.0	17.0
150 µ	70	14.0	97.0	3.0
Pan	15	3.0	—	—
Total	500	100	Σ = 278	

IS 383 grading limits for fine aggregate (percentage passing)

Sieve	Zone I	Zone II	Zone III	Zone IV	Sample (% passing)
10 mm	100	100	100	100	100
4.75 mm	90–100	90–100	90–100	95–100	97
2.36 mm	60–95	75–100	85–100	95–100	89
1.18 mm	30–70	55–90	75–100	90–100	71
600 µ	15–34	35–59	60–79	80–100	45
300 µ	5–20	8–30	12–40	15–50	17
150 µ	0–10	0–10	0–10	0–15	3

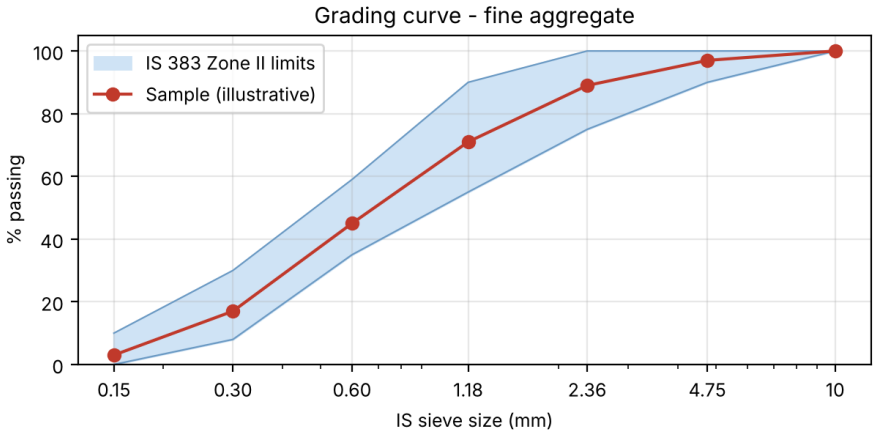


Fig. 2.1 Grading curve of the sample within Zone II limits

5. Calculations

% retained = (Mass retained ÷ Total mass) × 100; Cumulative % retained = running sum; % passing = 100 − cumulative % retained

Fineness Modulus (FM) = Σ (cumulative % retained on 10, 4.75, 2.36, 1.18, 0.6, 0.3, 0.15 mm sieves) ÷ 100

FM = (0 + 3 + 11 + 29 + 55 + 83 + 97) ÷ 100 = 278 ÷ 100 = 2.78

Grading zone: compare % passing with IS 383 Table 4 limits (below).

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Fineness modulus	2.78	Fine sand 2.2–2.6; medium 2.6–2.9; coarse 2.9–3.2
Grading zone	Zone II	All six sieves within Zone II limits
Suitability	Suitable for concrete	FM within 2.2–3.2

Result / Acceptance criteria

Fineness modulus of the sand = **2.78**; grading conforms to **Zone II** of IS 383 — suitable for use in concrete.

7. Precautions

- Do not overload the sieves (clogging); do not force particles through the mesh.
- Dry the sample completely; use sieves that are clean and undamaged.
- Record weights accurately to 0.1 g.

Experiment 2.2: Specific Gravity of Fine Aggregate (Pycnometer Method)

Course Outcome: CO2 | **Reference standard:** IS 2386 (Part 3)

1. Aim

To determine the specific gravity of fine aggregate, required for mix design and absolute-volume calculations.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Sample	≈ 200 g oven-dry sand (passing 10 mm sieve)
Pycnometer	1 litre wide-mouth jar with conical screw cap and 6 mm hole at apex
Balance	Accuracy 0.1 g
Others	Distilled water, glass rod, funnel, oven

3. Procedure

1. Weigh the clean, dry pycnometer (W_1); add about 200 g of dry sand and weigh again (W_2).
2. Fill with distilled water to the top, screw the cap and rotate/shake to remove entrapped air; weigh (W_3) after wiping the outside.
3. Empty, wash, fill the pycnometer with distilled water only to the same level and weigh (W_4).
4. Repeat the test and report the mean.

4. Observations — Weighings

Sample readings are illustrative; students must replace them with their own laboratory readings.

Observation	Symbol	Mass (g)
Empty pycnometer	W1	650.0
Pycnometer + dry sand	W2	850.0
Pycnometer + sand + water	W3	1934.5
Pycnometer + water	W4	1810.0

5. Calculations

Specific gravity $G = (W_2 - W_1) \div [(W_2 - W_1) - (W_3 - W_4)]$

$G = 200 \div [200 - (1934.5 - 1810.0)] = 200 \div 75.5 = 2.649 \approx 2.65$

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Specific gravity of sand	2.65	Typical 2.6–2.7 (natural sand)
Mean of two trials	2.65	Agreement within 0.02

Result / Acceptance criteria

Specific gravity of the fine aggregate = **2.65** (within the normal range for natural sand).

7. Precautions

- Remove all air bubbles by rotating the pycnometer gently.
- Outside of the pycnometer must be dry while weighing.
- Keep water temperature constant during the test.

Experiment 2.3: Water Absorption of Fine Aggregate

Course Outcome: CO2 | **Reference standard:** IS 2386 (Part 3)

1. Aim

To determine the water absorption of fine aggregate in the saturated-surface-dry (SSD) condition.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Sample	500 g sand passing 4.75 mm sieve
SSD test cone	Metal cone mould 40 mm top dia, 90 mm bottom dia, 75 mm high + tamper (340 g)
Other	Tray, absorbent cloth, oven (105–110 °C), balance (0.1 g), container

3. Procedure

1. Immerse about 500 g of sand in water for 24 h.
2. Drain the water, spread on a tray and dry with a warm gentle air current, stirring frequently until it flows freely.
3. Check SSD condition: fill cone loosely, tamp 25 times; remove the cone — the sand must just slump slightly (SSD). Weigh the SSD sand (W_1).
4. Oven-dry the sand at 105–110 °C for 24 h, cool and weigh (W_2).

4. Observations — Mass readings

Sample readings are illustrative; students must replace them with their own laboratory readings.

Mass of SSD sample, W1 (g)	Mass of oven-dry sample, W2 (g)	Water absorbed (W1 − W2) (g)
500.0	492.5	7.5

5. Calculations

Water absorption (%) = $[(W_1 - W_2) \div W_2] \times 100$

= (7.5 ÷ 492.5) × 100 = 1.52%

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Water absorption	1.52 %	Typical sand 0.5–2%
Free-water correction	Needed in mix design	Reduce mixing water by absorbed water

Result / Acceptance criteria

Water absorption of the sand = **1.52%**. Sand absorbing this much water must be taken into account in the water–cement ratio.

7. Precautions

- Do not over-dry; the sand should stay at the SSD point.
- Cool the sample in a desiccator before weighing.
- Avoid losing fines while handling.

Experiment 2.4: Bulking of Sand

Course Outcome: CO2 | **Reference standard:** IS 2386 (Part 3)

1. Aim

To determine the increase in volume (bulking) of fine aggregate due to moisture content and to find the moisture content for maximum bulking.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Sample	About 2 kg of dry sand; moisture additions of 2, 4, 6, 8, 10, 12% by weight
Measuring cylinder	250 mL or 500 mL graduated glass cylinder
Others	Balance, tray, scoop, steel rule, water

3. Procedure

1. Fill the measuring cylinder with loose sand to a height h_1 (100 mm / mark) without compaction.
2. Pour sand into a tray, add 2% water by weight, mix thoroughly and refill the cylinder loosely to the same mass; note height h_1 (bulked).
3. Pour this sand into a water-filled cylinder (or fill with water) and stir to expel air; note the inundated (saturated) height h_2.
4. Repeat for moisture contents 4, 6, ... 12% and plot bulking vs moisture.

4. Observations — Height readings (h1 = 100 mm for each moisture level)

Sample readings are illustrative; students must replace them with their own laboratory readings.

Moisture content (%)	h1 – loose moist sand (mm)	h2 – after inundation (mm)	Bulking (%) = (h1 − h2)/h2 × 100
0	100	100	0.0
2	100	89	12.4
4	100	83	20.5
6	100	80	25.0
8	100	82	22.0
10	100	85	17.6
12	100	88	13.6

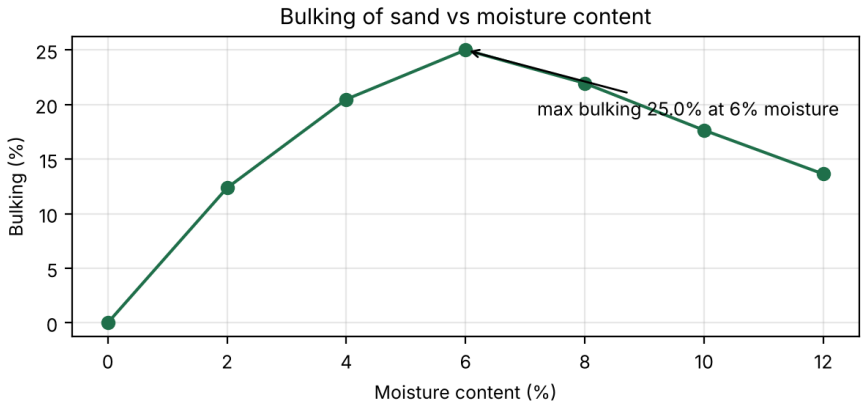


Fig. 2.4 Bulking curve of sand (illustrative)

5. Calculations

Bulking (%) = $[(h_1 - h_2) \div h_2] \times 100$

At 6% moisture: (100 − 80) ÷ 80 × 100 = 25%

Bulking factor = 1 + bulking/100 = 1.25 → volume of damp sand to be batched = 1.25 × dry volume

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Maximum bulking	25.0 %	Typically 20–30%
Moisture at maximum bulking	≈ 6 %	Usually 4–6%
Bulking factor	1.25	Volume correction for batching

Result / Acceptance criteria

Maximum bulking of the sand = **25%** at about **6% moisture content**; at about 12% moisture the sand volume reverts almost to the dry volume. For volume batching the sand quantity must be increased by the bulking percentage.

7. Precautions

- Do not compact the sand while filling the cylinder.
- Mix water thoroughly and uniformly into the sand.
- Take readings at eye level to avoid parallax.

Experiment 3.1: Grading of Coarse Aggregate (Sieve Analysis)

Course Outcome: CO2 | **Reference standard:** IS 2386 (Part 1); IS 383

1. Aim

To determine the particle-size distribution and fineness modulus of coarse aggregate and to check its grading against IS 383.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Sample	5 kg oven-dry 20 mm nominal size crushed aggregate
IS sieves	80, 40, 20, 16, 12.5, 10, 4.75 mm, 2.36 mm … 150 µ (FM series) and pan
Equipment	Sieve shaker, balance (1 g), trays, brush

3. Procedure

1. Dry the sample at 105–110 °C and weigh 5 kg.
2. Arrange sieves in descending order, place the sample on the top sieve and shake for 10 min.
3. Weigh the retained fraction on each sieve; clean the sieves by hand — do not force particles.
4. Check the sum of weights equals the initial weight within 0.5%.
5. Compute % retained, cumulative % retained, % passing and the fineness modulus.

4. Observations — Sieve analysis (sample mass = 5000 g)

Sample readings are illustrative; students must replace them with their own laboratory readings.

IS sieve	Mass retained (g)	% retained	Cumulative % retained	% passing
40 mm	0	0.0	0.0	100.0
20 mm	400	8.0	8.0	92.0
16 mm	1250	25.0	33.0	67.0
12.5 mm	1800	36.0	69.0	31.0
10 mm	1000	20.0	89.0	11.0
4.75 mm	480	9.6	98.6	1.4
Pan	70	1.4	—	—
Total	5000	100		

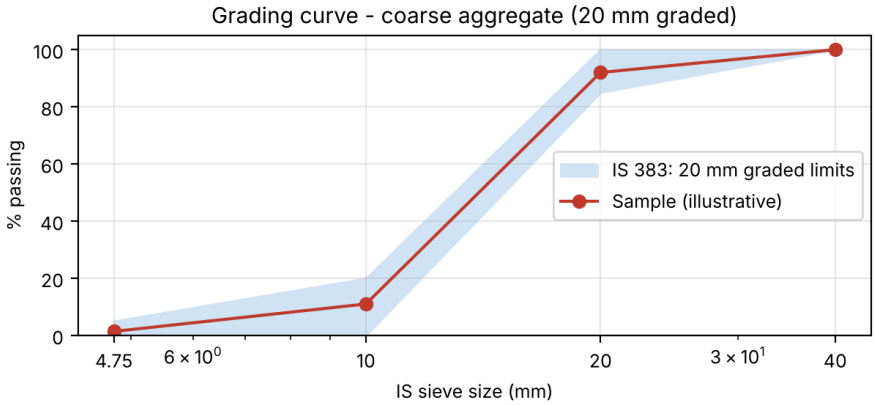


Fig. 3.1 Grading of the 20 mm coarse aggregate sample

5. Calculations

% passing = 100 − cumulative % retained. FM = Σ(cumulative % retained on 80, 40, 20, 10, 4.75, 2.36, 1.18 mm, 600, 300, 150 µ sieves) ÷ 100

All sieves finer than 4.75 mm retain 100% cumulatively: FM = (0 + 0 + 8.0 + 89.0 + 98.6 + 500) ÷ 100 = 5.96

IS 383 limits for 20 mm graded aggregate (% passing): 40 mm – 100; 20 mm – 85–100; 10 mm – 0–20; 4.75 mm – 0–5

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
% passing 20 mm	92.0 %	Limit 85–100 ✓
% passing 10 mm	11.0 %	Limit 0–20 ✓
% passing 4.75 mm	1.4 %	Limit 0–5 ✓
Fineness modulus	5.96	Typical 6.0–7.5

Result / Acceptance criteria

Coarse aggregate conforms to IS 383 for **20 mm nominal size graded aggregate**; fineness modulus = **5.96**.

7. Precautions

- Sample must be representative (use quartering/riffling).
- Do not overload sieves; clean each sieve thoroughly.
- Wear dust mask and gloves while shaking.

Experiment 3.2: Specific Gravity of Coarse Aggregate (Wire-Basket Method)

Course Outcome: CO2 | **Reference standard:** IS 2386 (Part 3)

1. Aim

To determine the specific gravity (and apparent specific gravity) of coarse aggregate for use in concrete mix design.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Sample	≈ 2 kg aggregate (> 10 mm), washed to remove dust
Wire basket	Perforated basket of ≈ 200 mm × 200 mm suspended from balance
Equipment	Balance with stirrup (1 g), water tank at 22–32 °C, soft cloth, oven

3. Procedure

1. Wash the sample, immerse in water for 24 h in the basket (tap to remove air).
2. Weigh basket + aggregate in water (B) and the empty basket in water (C).
3. Remove the aggregate, surface-dry with a cloth and weigh in air (A, SSD condition).
4. Oven-dry at 105–110 °C for 24 h, cool and weigh (D).

4. Observations — Weighings

Sample readings are illustrative; students must replace them with their own laboratory readings.

Symbol	Observation	Mass (g)
A	Saturated surface-dry sample in air	2000
B	Basket + sample in water	1635
C	Basket in water (empty)	375
B − C	Sample in water	1260
D	Oven-dry sample in air	1985

5. Calculations

Specific gravity (oven-dry basis) = D ÷ [A − (B − C)] = 1985 ÷ (2000 − 1260) = 1985 ÷ 740 = 2.68

Specific gravity (SSD basis) = A ÷ [A − (B − C)] = 2000 ÷ 740 = 2.70

Apparent specific gravity = D ÷ [D − (B − C)] = 1985 ÷ (1985 − 1260) = 1985 ÷ 725 = 2.74

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Specific gravity (oven-dry)	2.68	Typical 2.6–2.9
Specific gravity (SSD)	2.70	Used in mix design
Apparent specific gravity	2.74	—

Result / Acceptance criteria

Specific gravity of the coarse aggregate = **2.68** (oven-dry), **2.70** (SSD) and **2.74** (apparent).

7. Precautions

- Entrapped air must be removed by tapping the basket.
- Do not touch the wire basket sides while weighing in water.
- Keep the water temperature constant.

Experiment 3.3: Water Absorption of Coarse Aggregates

Course Outcome: CO2 | **Reference standard:** IS 2386 (Part 3)

1. Aim

To determine the water absorption of coarse aggregate, which affects the effective water–cement ratio.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Sample	≈ 2 kg aggregate (same as Exp. 3.2 or fresh sample)
Equipment	Balance (1 g), wire basket, oven, tray, absorbent cloth

3. Procedure

1. Immerse the washed aggregate in water for 24 h at 22–32 °C.
2. Remove, roll in a dry cloth until visible films disappear (SSD condition) and weigh (A).
3. Dry in an oven at 105–110 °C for 24 h, cool in a desiccator and weigh (D).

4. Observations — Weighings

Sample readings are illustrative; students must replace them with their own laboratory readings.

Mass of SSD sample, A (g)	Mass of oven-dry sample, D (g)	Water absorbed A − D (g)
2000	1985	15

5. Calculations

Water absorption (%) = [(A − D) ÷ D] × 100

= (15 ÷ 1985) × 100 = 0.76%

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Water absorption	0.76 %	Typical 0.1–2%; IS 383 permits up to 2% for ordinary use

Result / Acceptance criteria

Water absorption of the coarse aggregate = **0.76%**, i.e. a dense, low-porosity aggregate suitable for concrete.

7. Precautions

- Ensure complete saturation (24 h) before weighing.
- Dry the surface carefully — neither too wet nor too dry.
- Allow oven-dried sample to cool before weighing.

Experiment 4.1: Workability of Concrete — Compaction Factor Test

Course Outcome: CO3 | **Reference standard:** IS 1199

1. Aim

To determine the workability of concrete from the degree of compaction achieved under standard conditions; suitable for low and medium workability concrete (slump 10–50 mm).

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Mix	Nominal mix M20 (1 : 1.5 : 3, cement : sand : 20 mm aggregate by mass); w/c = 0.45 – 0.55 as stated
Batch (per trial)	Cement 5 kg, sand 7.5 kg, 20 mm aggregate 15 kg, water = w/c × 5 kg
Compacting factor apparatus	Two conical hoppers (upper 250/125 mm × 225 mm; lower 225/125 mm × 225 mm) with trap doors, cylinder 150 mm dia × 300 mm high, stand
Other	Balance (10 g), trowel, tamping rod, steel straight edge

3. Procedure

1. Weigh the empty cylinder (W_1) and place it on the base of the apparatus.
2. Fill the upper hopper gently with freshly mixed concrete without compaction; level the top.
3. Open the upper trap door; the concrete falls into the lower hopper. Open the lower trap door so that it falls into the cylinder.
4. Remove surplus concrete above the cylinder top with trowels and weigh the cylinder with partially compacted concrete (W_2).
5. Empty the cylinder, refill in layers, compact fully by tamping/vibration, level and weigh (W_3).
6. Repeat for different water–cement ratios.

4. Observations — Weighings and compaction factor (cylinder volume ≈ 5.30 litres)

Sample readings are illustrative; students must replace them with their own laboratory readings.

w/c	Empty cylinder W1 (kg)	Partly compacted W2 (kg)	Fully compacted W3 (kg)	Partial concrete W2 − W1 (kg)	Full concrete W3 − W1 (kg)	CF	Workability
0.45	6.5	17.70	19.20	11.2	12.7	0.882	Low
0.50	6.5	18.30	19.22	11.8	12.72	0.928	Medium
0.55	6.5	18.60	19.20	12.1	12.7	0.953	High

5. Calculations

Compaction factor (CF) = $(W_2 - W_1) \div (W_3 - W_1)$

w/c = 0.50: CF = 11.80 ÷ 12.72 = 0.928 ≈ 0.93

Workability classes (IS 456 / Shetty): very low 0.78; low 0.85; medium 0.92; high 0.95

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
CF at w/c 0.45	0.882	Low workability
CF at w/c 0.50	0.928	Medium workability
CF at w/c 0.55	0.953	High workability

Result / Acceptance criteria

Compaction factor of the given M20 mix at w/c = 0.50 is **0.93 → medium workability**; CF increases with water–cement ratio.

7. Precautions

- Fill the hoppers without any compaction or tapping.
- Concrete must be tested within 15–20 min of mixing.
- Clean the hoppers and cylinder after each test.

Experiment 4.2: Workability of Concrete — Slump Test

Course Outcome: CO3 | **Reference standard:** IS 1199; IS 7320

1. Aim

To determine the consistency (workability) of fresh concrete by the slump cone test; suitable for medium to high workability mixes.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Mix	Nominal mix M20 (1 : 1.5 : 3, cement : sand : 20 mm aggregate by mass); w/c = 0.45 – 0.55 as stated
Batch (per trial)	Cement 5 kg, sand 7.5 kg, 20 mm aggregate 15 kg, water = w/c × 5 kg
Slump cone	Frustum 300 mm high, 200 mm bottom dia, 100 mm top dia, steel 1.6 mm thick, with handles/foot pieces
Other	Smooth non-absorbent steel base plate, tamping rod 16 mm dia × 600 mm long (bullet end), steel scale, trowel

3. Procedure

1. Clean the cone and base plate, dampen and place the cone on the horizontal plate; hold with the foot pieces.
2. Fill the cone in three layers (each ≈ ⅓ of height); compact each layer with 25 strokes of the rod distributed uniformly (rod just penetrates previous layer).
3. Strike off the top surface with the tamping rod/trowel.
4. Lift the cone vertically upward slowly within 5–10 s without any lateral or torsional motion.
5. Measure the slump: difference between the height of the cone (300 mm) and the highest point of the subsided concrete; note the type of slump (true/shear/collapse).

4. Observations — Slump observations

Sample readings are illustrative; students must replace them with their own laboratory readings.

w/c	Initial height (mm)	Height after lifting cone (mm)	Slump (mm)	Type of slump	Workability
0.45	300	275	25	True	Very low–Low
0.50	300	240	60	True	Medium
0.55	300	195	105	True	High

5. Calculations

Slump (mm) = Height of mould (300 mm) − Height of subsided concrete (mm)

w/c = 0.50: Slump = 300 − 240 = 60 mm

Degrees of workability (Neville/Shetty): very low 0–25 mm; low 25–50 mm; medium 50–100 mm; high 100–175 mm

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Slump at w/c 0.45	25 mm	Very low / low
Slump at w/c 0.50	60 mm	Medium — suitable for RCC beams/slabs
Slump at w/c 0.55	105 mm	High — suitable for congested sections

Result / Acceptance criteria

Slump of the M20 mix at w/c = 0.50 is **60 mm** (true slump) → medium workability. A *shear* slump indicates a harsh/lean mix; a *collapse* indicates an excessively wet mix.

7. Precautions

- Place the cone on a flat, non-absorbent and non-vibrating base.
- Lift the cone vertically; complete the test within 2½ min of sampling.
- Do not remeasure a shear slump — repeat the test with fresh concrete.

Experiment 4.3: Workability of Concrete — Vee-Bee Test

Course Outcome: CO3 | **Reference standard:** IS 1199 (Vee-Bee consistometer)

1. Aim

To determine the workability of stiff / very low workability concrete by measuring the time required for remoulding under vibration.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Mix	Nominal mix M20 (1 : 1.5 : 3, cement : sand : 20 mm aggregate by mass); w/c = 0.45 – 0.55 as stated
Batch (per trial)	Cement 5 kg, sand 7.5 kg, 20 mm aggregate 15 kg, water = w/c × 5 kg
Vee-Bee consistometer	Vibrating table 380 × 260 mm (50 Hz), cylindrical container 240 mm dia × 200 mm high, swivel arm with funnel & glass disc (230 mm dia, 10 mm thick)
Other	Slump cone, tamping rod, stop-watch, trowel

3. Procedure

1. Place the slump cone in the cylindrical container fixed to the vibrating table; fill with concrete in 3 layers, each tamped with 25 strokes.
2. Remove the cone by lifting vertically; swing the glass disc attachment over the concrete and lower it to touch the top.
3. Switch on the vibrator and start the stop-watch simultaneously.
4. Observe the concrete through the transparent disc; stop the watch the moment the whole surface of the disc is covered with cement paste (remoulding complete).

4. Observations — Vee-Bee time

Sample readings are illustrative; students must replace them with their own laboratory readings.

w/c	Vee-Bee time (s)	Workability description
0.40	14	Very stiff
0.45	8	Stiff
0.50	4	Stiff plastic

5. Calculations

Vee-Bee degree = time in seconds for the glass disc to be fully covered with mortar

Classes (Shetty): extremely dry 32–18 s; very stiff 18–10 s; stiff 10–5 s; stiff plastic 5–3 s; plastic 3–0 s

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Vee-Bee time at w/c 0.40	14 s	Very stiff
Vee-Bee time at w/c 0.45	8 s	Stiff
Vee-Bee time at w/c 0.50	4 s	Stiff plastic

Result / Acceptance criteria

Vee-Bee time of the mix at w/c = 0.50 is **4 s** (stiff plastic). The Vee-Bee test is suitable for dry concrete where slump gives no meaningful reading.

7. Precautions

- Do not lift the cone with a jerk.
- Stop the watch exactly when the disc surface is wholly covered by paste.
- Clean the container and disc immediately after testing.

Experiment 5.1: Compressive Strength of Concrete (Cube Test)

Course Outcome: CO4 | **Reference standard:** IS 516; IS 456 (acceptance)

1. Aim

To determine the compressive strength of concrete cubes at 7 and 28 days and to compare with the characteristic strength of the grade (M20).

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Mix	Nominal mix M20 (1 : 1.5 : 3, cement : sand : 20 mm aggregate by mass); w/c = 0.45 – 0.55 as stated
Specimens	150 mm cubes: 3 cubes for 7 days + 3 cubes for 28 days (cube area = 22 500 mm²)
Moulds	Cast iron / steel 150 × 150 × 150 mm, tight, oiled
Equipment	Tamping rod 16 mm dia, vibrating table (optional), curing tank at 27 ± 2 °C
Testing machine	Compression testing machine (CTM) 2000 kN; load rate 14 N/mm² per min (≈ 315 kN/min)

3. Procedure

1. Mix concrete and fill the moulds in three layers, compacting each with 35 strokes of the tamping rod (or vibrate); level and finish the top.
2. Cover the moulds and store for 24 ± 0.5 h at 27 ± 2 °C and ≥ 90% RH; demould and mark the cubes.
3. Cure the cubes in clean water at 27 ± 2 °C until the age of testing; remove just before testing and wipe off surface water.
4. Place the cube centrally in the CTM with the cast faces at the sides (not top and bottom).
5. Apply the load without shock at 14 N/mm² per min until the cube fails; record the maximum load.

4. Observations — Failure loads and strength (kN → N/mm² on 22 500 mm²)

Sample readings are illustrative; students must replace them with their own laboratory readings.

Age (days)	Cube 1	Cube 2	Cube 3	Average (N/mm²)
7	390 kN → 17.33	400 kN → 17.78	385 kN → 17.11	17.41
28	600 kN → 26.67	615 kN → 27.33	605 kN → 26.89	26.96

5. Calculations

Compressive strength f_c = Failure load P (N) ÷ Cross-sectional area A (mm²) = P ÷ 22 500

28-day average = (600, 615, 605 kN → 26.96 N/mm²)

Strength ratio 7-day / 28-day = 17.41 ÷ 26.96 = 0.65 (≈ 0.65 expected)

Acceptance (IS 456, cl. 16): mean of 4 non-overlapping consecutive test results ≥ f_{ck} + 0.825 s, and each result ≥ f_{ck} − 3

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
7-day compressive strength	17.41 N/mm²	≈ 65% of 28-day (65%)
28-day compressive strength	26.96 N/mm²	Target mean strength for M20 ≈ 26.6 N/mm²
Grade achieved	M20 satisfied	26.96 > f_{ck} = 20 N/mm²

Result / Acceptance criteria

28-day compressive strength of the concrete = **26.96 N/mm²** (> 20 N/mm²) — **M20 grade is satisfied**.

7. Precautions

- Moulds must be tight and well oiled; fill and compact in proper layers.
- Do not allow cubes to dry before testing; test in surface-dry condition.
- Apply the load at the specified rate and without shock.

Experiment 5.2: Modulus of Rupture (Flexural Strength) of Concrete

Course Outcome: CO4 | **Reference standard:** IS 516 (two-point loading)

1. Aim

To determine the flexural tensile strength (modulus of rupture) of concrete beams under two-point (third-point) loading.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Mix	Nominal mix M20 (1 : 1.5 : 3, cement : sand : 20 mm aggregate by mass); w/c = 0.45 – 0.55 as stated
Specimens	3 beams 150 × 150 × 700 mm cured 28 days
Flexure testing machine	UTM/flexure frame with two loading rollers and two support rollers (38 mm dia); span 600 mm; loading points 200 mm apart (L/3)
Loading rate	0.7 N/mm² per min (≈ 4 kN/min for a 150 mm beam)

3. Procedure

1. Take the beam from the curing tank, wipe it dry, and mark the supports and load points; the casting-side faces must be the loading faces (rotate beam 90° from cast position).
2. Place the beam on the supports (span 600 mm) with the loading rollers 200 mm apart and symmetrically positioned.
3. Apply load without shock at the specified rate until the beam fractures; record the maximum load P.
4. Measure the distance ‘a’ from the fracture line to the nearer support (along the tension face).

4. Observations — Failure loads and flexural strength (L = 600 mm, b = d = 150 mm)

Sample readings are illustrative; students must replace them with their own laboratory readings.

Beam	Failure load P (kN)	Fracture distance ‘a’ from nearer support (mm)	Location of crack	f_b (N/mm²)
1	17.0	215	Middle third	3.02
2	18.2	205	Middle third	3.24
3	17.6	225	Middle third	3.13
Mean				3.13

5. Calculations

If the crack is in the middle third (a > 200 mm): $f_b = P \times L \div (b \times d^2)$

If the crack is outside the middle third but a ≥ 170 mm: $f_b = 3\,P\,a \div (b \times d^2)$ (discard if a < 170 mm)

Beam 1: $f_b = 17\,000 \times 600 \div (150 \times 150^2) = 3.02$ N/mm²; Mean = 3.13 N/mm²

Empirical check (IS 456): $f_{cr} = 0.7\sqrt{f_{ck}} = 0.7 \times \sqrt{20} = 3.13$ N/mm²

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Mean modulus of rupture	3.13 N/mm²	IS 456: $0.7\sqrt{f_{ck}} = 3.13$ N/mm²
Ratio to cube strength	≈ 11–12%	Flexure strength typically 10–15% of f_c

Result / Acceptance criteria

Modulus of rupture of the concrete = **3.13 N/mm²**, in agreement with the code value of 3.13 N/mm² for M20.

7. Precautions

- Ensure the beam is aligned and symmetrically placed on the rollers.
- Apply load slowly and uniformly — impact loads give wrong results.
- Dispose of fractured specimens safely.

Experiment 5.3: Young's Modulus and Poisson's Ratio of Concrete

Course Outcome: CO4 | **Reference standard:** IS 516 (static modulus of elasticity); IS 456 cl. 6.2.3.1

1. Aim

To determine the static modulus of elasticity (Young's modulus, E_c) and Poisson's ratio (μ) of concrete from the stress–strain behaviour of a cylinder in uniform compression.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Specimens	150 mm dia × 300 mm cylinders (area = 17 671 mm²), cured 28 days; one is used for ultimate load
Strain measurement	Compressometer (gauge length 200 mm) for axial strain + extensometer / strain gauge for lateral strain
Testing machine	CTM 2000 kN; load rate ≈ 0.25 N/mm² per s
Capping	Cement paste/sulphur capping for plane, parallel ends

3. Procedure

1. Test one cylinder to failure to find the ultimate strength (f_c); the working load is taken as 1/3 of this value (stress ≈ 0.33 f_c).
2. Fix the compressometer and extensometer on a second cylinder, centred in the CTM.
3. Apply load in increments of 40 kN up to 1/3 of the ultimate; record axial and lateral strain at each increment.
4. Unload and repeat 2–3 cycles; use the final cycle for calculations (the stress–strain graph should be nearly linear).
5. Plot stress vs axial strain (slope = E) and lateral strain vs axial strain (slope = μ).

4. Observations — Load, stress and strain readings

Sample readings are illustrative; students must replace them with their own laboratory readings.

Load (kN)	Stress σ (N/mm²)	Axial strain ε_a (×10⁻⁶)	Lateral strain ε_l (×10⁻⁶)
0	0.00	0	0.0
40	2.26	91	15.4
80	4.53	181	30.8
120	6.79	272	46.2
160	9.05	362	61.6

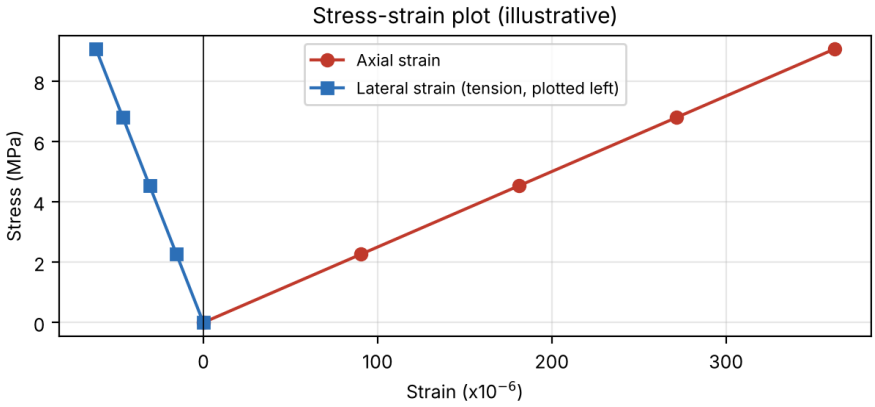


Fig. 5.3 Stress–strain plot (axial strain to the right, lateral strain to the left)

5. Calculations

Stress σ = Load ÷ Area = P ÷ 17 671

Young's modulus $E_c = \Delta\sigma \div \Delta\varepsilon_a = (9.05 - 0) \div (362 \times 10^{-6}) = 25{,}000$ N/mm²

Poisson's ratio μ = Lateral strain ÷ Axial strain = 61.6 ÷ 362 = 0.17

IS 456 check: $E_c = 5000\sqrt{f_{ck}} = 5000\sqrt{20} = 22\,360$ N/mm² (for M20)

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Young’s modulus	25,000 N/mm²	IS 456: 22 360 N/mm² (M20); 25 000 for M25
Poisson’s ratio	0.17	Typical range 0.15–0.20

Result / Acceptance criteria

Static modulus of elasticity of the concrete = **25,000 N/mm²** and Poisson’s ratio = **0.17**.

7. Precautions

- Centre the specimen so that the load is axial; mount gauges rigidly.
- Do not exceed one-third of ultimate load during the modulus test.
- Remove the strain gauges before loading to failure to avoid damage.

Experiment 5.4: Split Tensile Strength of Concrete

Course Outcome: CO4 | **Reference standard:** IS 5816

1. Aim

To determine the indirect tensile strength of concrete by splitting a cylinder along its vertical diameter (Brazilian test).

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Specimens	3 cylinders 150 mm dia × 300 mm long, cured 28 days
Packing strips	Plywood strips ≈ 3 mm thick × 15 mm wide × 300 mm long (use once)
Testing machine	CTM; splitting-stress rate 1.2–2.4 N/mm² per min (≈ 85–170 kN/min for 150 × 300 cylinder)
Other	Marking gauge, vernier calipers, balance

3. Procedure

1. Draw diametrical lines on both ends of the cylinder; measure dia D and length L.
2. Place a plywood strip on the lower plate, place the cylinder horizontally with the marked lines vertical, and place the second strip on top.
3. Bring the upper platen down to touch the strip; apply load without shock at the specified rate until the cylinder splits.
4. Record the failure load P.

4. Observations — Failure loads

Sample readings are illustrative; students must replace them with their own laboratory readings.

Cylinder	D (mm)	L (mm)	Failure load P (kN)	Split tensile strength f_{ct} (N/mm²)
1	150	300	165	2.33
2	150	300	175	2.48
3	150	300	170	2.41
Mean				2.41

5. Calculations

$f_{ct} = 2\,P \div (\pi\,L\,D)$

Cylinder 1: $f_{ct} = 2 \times 165\,000 \div (\pi \times 300 \times 150) = 2.33$ N/mm²

Mean $f_{ct} = 2.41$ N/mm²; ratio $f_{ct}/f_c = 2.41 \div 26.96 \approx 9\%$

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Mean split tensile strength	2.41 N/mm²	Typically 8–12% of compressive strength
Compared with flexural strength	3.13 N/mm²	Split tensile is ≈ 0.7–0.8 × modulus of rupture

Result / Acceptance criteria

Split tensile strength of the concrete = **2.41 N/mm²**.

7. Precautions

- Place packing strips and cylinder symmetrically; the load line must pass through the diametrical plane.
- Apply load slowly and smoothly.
- Use fresh plywood strips for every test.

Experiment 5.5: Non-Destructive Testing of Concrete (Demonstration)

Course Outcome: CO4 | **Reference standard:** IS 13311 (Part 1) – UPV; IS 13311 (Part 2) – Rebound hammer

1. Aim

To assess the in-situ quality and approximate compressive strength of hardened concrete without damaging the member, using the rebound hammer and the ultrasonic pulse velocity (UPV) test.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Rebound (Schmidt) hammer	Impact energy 2.25 N·m (Type N); calibration anvil; correlation chart
UPV tester	Pulse generator/receiver, 54 kHz transducers, couplant (grease/jelly), calibration bar
Test member	Concrete cube/slab/beam of known age; clean, smooth surface
Others	Marking chalk, measuring tape, grinding stone

3. Procedure

1. **Rebound hammer:** smooth the test surface, mark a 150 × 150 mm grid with points at least 20 mm apart; press the plunger perpendicular to the surface until the mass releases.
2. Record the rebound number at each point (minimum 9–12 readings); discard readings differing from the average by more than 6 units and re-average.
3. Convert the rebound number to compressive strength with the correlation chart (corrected for hammer direction).
4. **UPV:** calibrate the instrument on the reference bar; apply couplant, place the transducers on opposite faces (direct method), and read the transit time t.
5. Compute pulse velocity V = L ÷ t and classify concrete quality per IS 13311 (Part 1).

4. Observations — Rebound hammer and UPV readings

Sample readings are illustrative; students must replace them with their own laboratory readings.

Reading no.	1	2	3	4	5	6	7	8	9	10	Mean
Rebound no.	32	34	33	35	31	33	34	32	33	36	33.3

Path length L (mm)	Transit time t (μs)	Pulse velocity V = L/t (km/s)	Quality of concrete
150	36	4.17	Good
150	33	4.55	Excellent

5. Calculations

Average rebound number = Σ ÷ n = 333 ÷ 10 = 33.3 (no reading deviates by more than 6 units → all accepted)

From the correlation chart (horizontal impact, hammer calibration): rebound number ≈ 33 ↔ compressive strength ≈ 27 N/mm²

UPV: V = L ÷ t = 150 mm ÷ 36 μs = 4.17 km/s

IS 13311 (Part 1) quality: > 4.5 km/s – excellent; 3.5–4.5 – good; 3.0–3.5 – medium; < 3.0 – doubtful

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Average rebound number	33.3	Corresponds to ≈ 27 N/mm² (chart)
Pulse velocity	4.17 – 4.55 km/s	Good → excellent
Quality assessment	Sound, uniform, no cracks / voids	Consistent with 28-day cube strength of Exp. 5.1

Result / Acceptance criteria

Average rebound number = **33.3** (≈ 27 N/mm²) and UPV = **4.17–4.55 km/s** — the concrete is of **good to excellent quality**. NDT results are indicative; cores/cubes give definitive strength.

7. Precautions

- Rebound hammer must be held perpendicular to the surface; avoid testing over rebar, honeycomb or rough/wet surfaces.
- Use a thick couplant film for UPV so that there is no air gap; avoid paths along reinforcement.
- Calibrate both instruments before use.

Appendix A: Summary of Experiments and Acceptance Limits

Exp.	Test	Key input	Key output (sample)	Acceptance limit
1.1	Normal consistency	300 g cement; w% 26→	P = 30%	Plunger 5–7 mm from bottom
1.2	Fineness	100 g cement; 90 µ	7.7% residue	≤ 10% (OPC)
1.3	Setting time	300 g; 0.85 P water	Initial 120 min; final 270 min	Initial ≥ 30; final ≤ 600 min
1.4	Specific gravity	64 g cement; kerosene	3.15	3.10–3.15
1.5	Soundness	100 g; 0.78 P water	2.25 mm	≤ 10 mm
1.6	Cement strength	200 g cement + 600 g sand	25.0 / 34.4 / 45.0 N/mm²	43 grade: 23 / 33 / 43
2.1	FA sieve analysis	500 g sand	FM 2.78, Zone II	FM 2.2–3.2; IS 383 zones
2.2	FA specific gravity	200 g sand	2.65	2.6–2.7
2.3	FA water absorption	500 g SSD sand	1.52%	≈ 0.5–2%
2.4	Bulking	Moisture 0–12%	25% at 6%	Max 20–30%
3.1	CA sieve analysis	5 kg 20 mm	FM 5.96	IS 383 graded 20 mm
3.2	CA specific gravity	2 kg	2.68	2.6–2.9
3.3	CA water absorption	2 kg	0.76%	≤ 2%
4.1	Compaction factor	M20; w/c 0.5	0.93	0.78–0.95 (very low–high)
4.2	Slump	M20; w/c 0.5	60 mm	As per placing conditions (IS 456 cl. 7.1)
4.3	Vee-Bee	M20; w/c 0.5	4 s	Stiff plastic 3–5 s
5.1	Cube strength	150 mm cubes	26.96 N/mm² (28 d)	≥ f_{ck} (20)
5.2	Flexural strength	150×150×700 beams	3.13 N/mm²	$0.7\sqrt{f_{ck}}$ = 3.13
5.3	E and µ	150×300 cylinder	E = 25,000 N/mm²; µ = 0.17	$5000\sqrt{f_{ck}}$; µ 0.15–0.20
5.4	Split tensile	150×300 cylinder	2.41 N/mm²	≈ 8–12% of f_c
5.5	NDT	Rebound hammer, UPV	R = 33.3; V = 4.2–4.5 km/s	V > 3.5 km/s = good

General laboratory safety

- Wear safety shoes, gloves, goggles and dust mask; cement is alkaline and may cause skin burns — wash immediately with water.
- Handle hot water baths, boiling Le Chatelier moulds and ovens with tongs/heat-resistant gloves.
- Keep kerosene away from flames; store in closed containers.
- Never stand in front of the cube/cylinder while it is being crushed in the CTM; use the safety guard.
- Calibrate balances, sieves, moulds and testing machines periodically; record the calibration date in the lab register.

Note: acceptance limits are quoted from commonly used BIS codes (IS 269, IS 383, IS 456, IS 516, IS 1199, IS 2386, IS 4031, IS 5816, IS 13311). Always confirm values against the latest code edition and your laboratory manual issued by the University / Department.